

Nathan Zhang

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EDUCATION

The University of British Columbia

B.A. in Economics & Statistics; Minor in Data Science

Vancouver, BC
Sep 2023 – May 2028

- **Cumulative Average:** 93.5 / 100
- **Awards:** Worklearn International Undergraduate Research Award (2026); Economics Undergraduate Scholarship (2025); Arts International Student Scholarship (2024, 2025)
- **Relevant Coursework:** Advanced Econometrics (98); Time Series & Forecasting (95); Machine Learning & Data Mining (93); Computational Optimization (92); Causal Methods (98); Probability (91); Applied Linear Algebra (100)

WORK & RESEARCH EXPERIENCE

UBC Sauder School of Business | ML/Finance/Econometrics – Dr. Allen Hu

Research Assistant

Vancouver, BC
May 2026 – Present

- Built a Jupyter-based workflow to construct post-2010 financial media corpora, screening major U.S. outlets for full-text access, missingness, metadata reliability, and sentiment-signal readiness; extracting text topics using LDA methods.
- Assessed unstructured text, video, and PDF sources for investment-research viability, documenting signal relevance and integration paths for downstream LLM applications; exploring causal discovery methods on unstructured sequence data.

UBC Sauder School of Business | Quant Labour/Econometrics – Dr. J. Frank Li

Research Assistant (Co-op)

Vancouver, BC
Apr 2026 – Present

- Built DuckDB and Polars parquet pipelines linking PSID worker histories, O*NET labour metrics, and USPTO patent features through specialized crosswalk logic, delivering regression-ready longitudinal panels and data validations.
- Deployed high-throughput Transformer similarity workflows on UBC HPC GPU nodes to map 8.5M patent records to occupational tasks, creating interpretable, lag-aware technology-exposure measures.
- Converted a theoretical employment-sorting model into measurable empirical objects by auditing worker-occupation matching, parameter interpretation, and regression design; drafted R/Jupyter replication package for publication.

Vancouver School of Economics at UBC | Econometrics/ML – Dr. Jonathan Graves

Research Assistant

Vancouver, BC
Jan 2025 – May 2026

- Developed reproducible Python/R workflows for panel econometrics, time-series forecasting, computer vision, text embeddings, and zero-shot classification for COMET and Praxis UBC research and teaching resources platforms.
- Maintained Quarto and GitHub Actions deployment pipelines for research materials, documenting assumptions, data flows, and code paths; supported online Jupyter image access for 1,000+ student users.

SELECTED RESEARCH & DATA PROJECTS

Double Machine Learning for Payment-Friction Spending Effects [Report]

Nov 2025 – Dec 2025

- Designed a DoubleML framework in Python (pandas, scikit-learn, XGBoost) on a 22k person-day panel to estimate the spending effect of low-friction payments while controlling for high-dimensional behavioral confounding.
- Applied cross-fitting, tree-based nuisance models and orthogonal scores; estimated a 5% spending increase and profiled A/B-testing-style treatment heterogeneity across consumer distributions.

Canadian Inflation Forecasting and Macro Signal Evaluation [Presentation]

Aug 2025

- Built an end-to-end R VAR forecasting pipeline on Canadian macro-financial indicators from CANSIM, combining stationarity diagnostics & transformations, AIC-driven lag selections, benchmark comparisons, and rolling-window validation.
- Achieved 0.16% RMSE in 12-month rolling backtests and interpreted impulse responses to translate statistical forecasts into policy-facing macro narratives and model-risk assessments.

Impact of Minimum Wage Policy on Education Choice, Econometrics Game [Paper]

Feb 2026

- Built a geospatial panel linking 1,293 U.S. counties with ED Facts records, using border-pair DiD exposure and fuzzy RDD designs to isolate local variation around state minimum-wage changes, reinforcing causal assumptions.
- Estimated a 2.3% absenteeism increase per dollar of minimum-wage hike with fixed-effects specifications, stress-testing results against state confounding and spatial comparability to confirm robustness.

TECHNICAL SKILLS

Languages & Tools: Python, R, SQL (Postgre, MySQL), Julia, MATLAB, Git, Docker, Unix/HPC, Tableau, Spark, Quarto

ML & Data: PyTorch, TensorFlow, Transformers, scikit-learn, pandas, NumPy, tidyverse, DuckDB, Polars, fixest, OpenCV

Quant Methods: Causal Inference (DiD, IV, RDD, Double ML), Ensemble Methods, Bayesian Methods, VAR, A/B testing